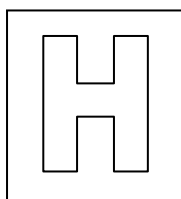


Candidate Name: _____

Class Adm No

--	--



2021 End of Year Exams Pre-university 2

BIOLOGY

8876/01

Paper 1

21 September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, Adm No. and class on all the papers you hand in.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions in this section. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Multiple Choice Answer Sheet.

This question paper consists of 22 printed pages.

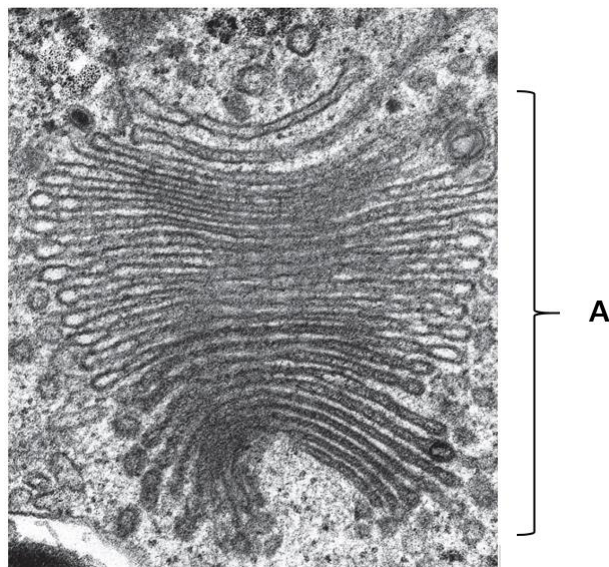
[Turn over

Answer **all** questions in this section.

1. How many layers of **phospholipids** will a molecule of water in the extracellular fluid have to pass to reach the thylakoid lumen?

A 2
B 4
C 6
D 8

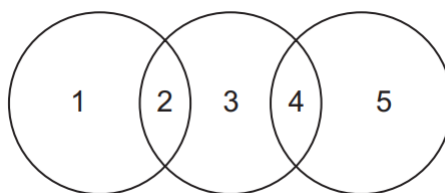
2. The following diagram shows an electron micrograph of a pancreatic cell.



Which of the following is **false** about organelle **A**?

- A The flattened sacs of membranes are not interconnected.
B Large amounts of ribosomes are embedded on the membranes of the flattened sacs.
C The flattened sacs of membranes allows for a high surface area to volume ratio to increase efficiency of the organelle.
D Proteins can be chemically modified by processes such as glycosylation in the lumen of the flattened sacs.

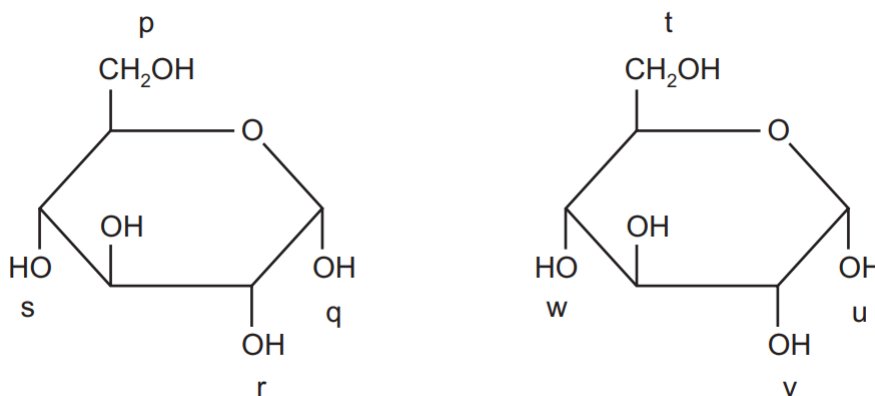
3. The diagram shows some similarities between chloroplasts, mitochondria and typical prokaryotes.



Which row is correct?

	1	2	3	4	5
A	chloroplasts	circular DNA	mitochondria	80S ribosomes	prokaryotes
B	chloroplasts	80S ribosomes	mitochondria	circular DNA	prokaryotes
C	prokaryotes	circular DNA	mitochondria	circular DNA	mitochondria
D	prokaryotes	70S ribosomes	chloroplasts	80S ribosomes	mitochondria

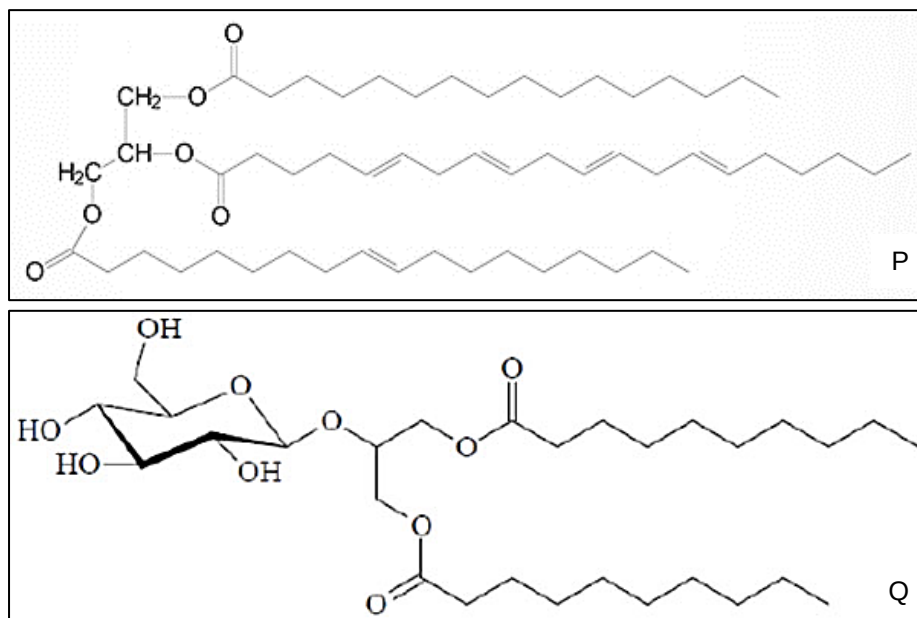
4. The diagram shows two molecules of glucose. Four possible bonding positions are labelled p, q, r and s and t, u, v, w.



When these two molecules condense to form glycogen, where could bonds form?

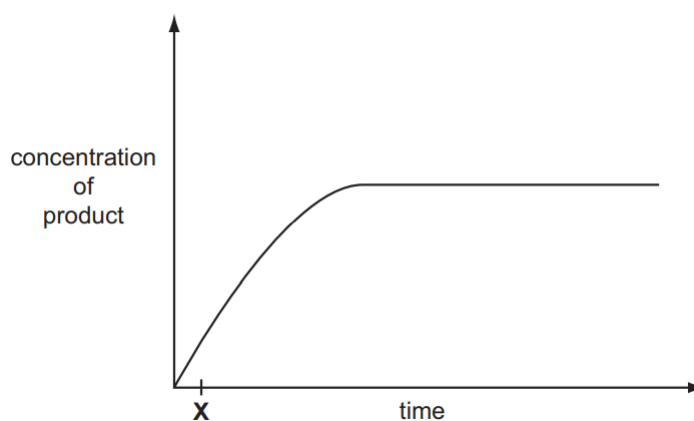
- A** p – u or p – v
B p – u or q – w
C p – v or q – w
D p – w or v – w

5. The following figure shows a triglyceride, P and a glycolipid, Q.



Which of the following statement is true?

- A Both P and Q are found in membranes.
 - B P consists of three ester bonds, while Q consists of three glycosidic bonds.
 - C Upon hydrolysis, Q will produce more soluble products than P.
 - D Both P and Q are good long-term energy store in hibernating animals.
6. The graph shows the course of an enzyme-catalysed reaction at 30°C.



What is true at time X?

- A Most enzyme molecules will have free active sites.
- B The number of available substrate molecules is high.
- C The number of enzyme-substrate complexes is low.
- D The rate remains the same if more enzyme is added.

7. The following are statements about ribozymes:

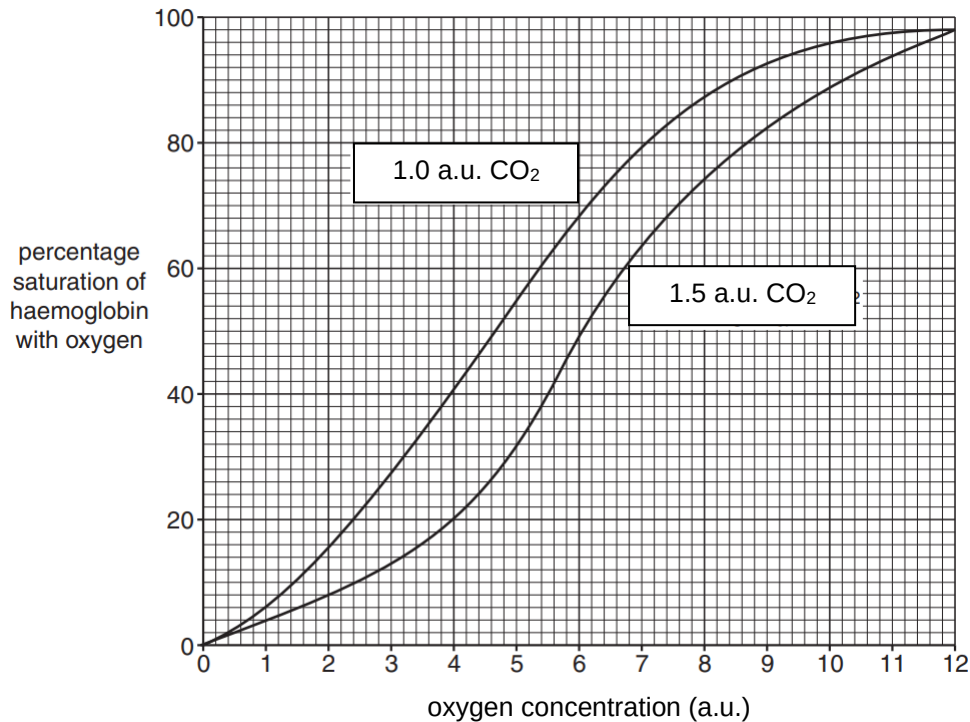
- It is an RNA molecule
- It is able to catalyse reactions in a similar way to protein enzymes
- It is hypothesised to be the first type of biological catalyst

Which of the following can be concluded from the information given?

- I It is formed via transcription.
- II It has a 3D conformation stabilized by various R-group interactions.
- III It has an active site that has a complementary shape to its substrates.
- IV It should be able to catalyse its own synthesis.

- A** I and III only
- B** II and IV only
- C** I, II and III only
- D** I, III and IV only

8. The following figure shows the effect of increasing carbon dioxide concentration on the oxygen haemoglobin dissociation curve. The oxygen haemoglobin dissociation curves shows the relationship between how much oxygen is bound to haemoglobin (percentage saturation of haemoglobin with oxygen) and oxygen concentration.



Based on the figure, which of the following statements are true?

- I When concentration of CO₂ is higher, haemoglobin binds oxygen less readily.
 - II When concentration of CO₂ is lower, the cooperative binding effect of haemoglobin is inhibited.
 - III At high CO₂ concentration, haemoglobin releases O₂ more readily.
 - IV At very high oxygen concentrations, CO₂ concentration has no effect on percentage saturation of haemoglobin with oxygen.
- A** I and IV only
B II and III only
C I, II and IV only
D I, III and IV only

9. Which of the following structural feature(s) of DNA enables it to replicate semi-conservatively?

- I complementary base pairing occurs between purines and pyrimidines.
- II double-stranded, where both strands act as templates.
- III deoxyribonucleotides are joined by covalent phosphodiester bonds.
- IV helical structure allows it to be packed in a compact manner.

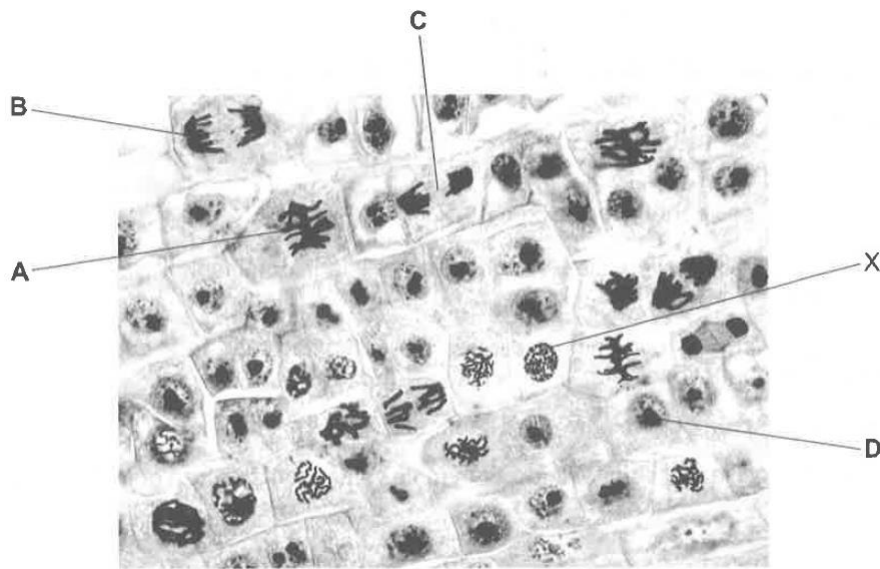
- A II only
- B I and II only
- C III and IV only
- D I, II and III only

10. A double-stranded DNA containing 4 800 nucleotides is transcribed and translated into a polypeptide consisting of amino acids of twelve different kinds.

What is the total number of amino acids used and the theoretical minimum number of different tRNA molecules required to translate the mRNA for this polypeptide?

- A 800 amino acids and 20 different tRNAs
- B 800 amino acids and 12 different tRNAs
- C 1 600 amino acids and 20 different tRNAs
- D 1 600 amino acids and 12 different tRNAs

11. The photomicrograph shows a plant tissue where mitosis is occurring.



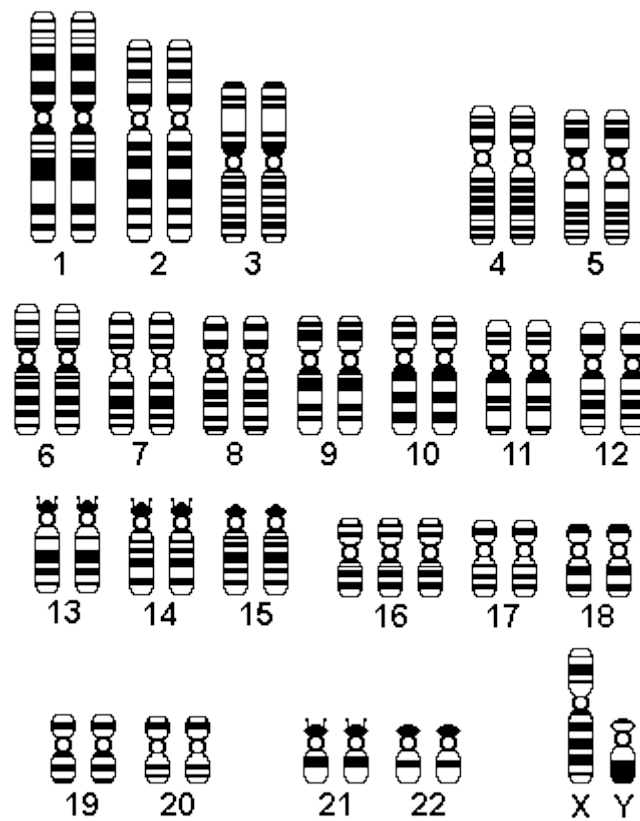
Which of the following shows the correct sequence of events?

- A C > B > A > X > D
- B D > X > B > A > C
- C X > A > B > C > D
- D X > C > A > B > D

12. Which of the following is **false** about sickle cell disease?

- A The total length of β -globin chain is unaffected.
- B Carriers of the disease have a mixture of normal haemoglobin and abnormal haemoglobin.
- C It is caused by a nonsense mutation, resulting in a valine being coded for instead of glutamic acid.
- D It is a recessive disease, as normal β -globin chain can still be produced with one normal allele present.

13. The following diagram shows one type of chromosomal abnormality.



Which of the following correctly identifies the type of chromosomal abnormality and a possible cause?

	type of chromosomal abnormality	possible cause
A	aneuploidy	error during meiosis I of father's germ cell, resulting in $n+1$ gamete
B	polyploidy	error during meiosis I of father's germ cell, resulting in $n+1$ gamete
C	aneuploidy	error during meiosis II of mother's germ cell, resulting in $n-1$ gamete
D	polyploidy	error during meiosis II of mother's germ cell, resulting in $n-1$ gamete

14. Cancer cells divide out of control, forming tumours.

Which statement describes the difference between a cancer cell and a normal cell?

- A** Cancer cells have a shorter interphase.
- B** Cancer cells are able to skip mitosis and enter cytokinesis immediately.
- C** Only cancer cells have mutated DNA.
- D** Cancer cells undergo more cytokinesis in one round of cell cycle.

15. Male bees are haploid. They develop from unfertilised eggs. Female bees are diploid.

Which statements are correct?

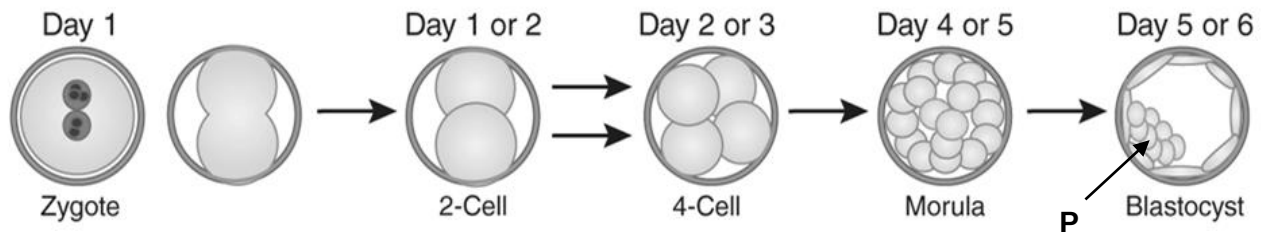
- 1 All male bees are genetically identical.
- 2 Male bee sperm cells are produced by mitosis.
- 3 Only gametes formed by female bees can contain new combinations of alleles.

- A** 1 and 2 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 1, 2 and 3

16. Which pair of statements correctly describes how cellular DNA content and ploidy level change after meiosis I and meiosis II?

- A** Statement 1: DNA content per cell is halved after both meiosis I and meiosis II.
Statement 2: Ploidy level remains diploid after meiosis I, and halve to haploid after meiosis II.
- B** Statement 1: DNA content per cell is halved after both meiosis I and meiosis II.
Statement 2: Ploidy level is halved from diploid to haploid after meiosis I and remains haploid after meiosis II.
- C** Statement 1: DNA content per cell is halved after meiosis I and remains unchanged after meiosis II.
Statement 2: Ploidy level is halved from diploid to haploid after meiosis I and remains haploid after meiosis II.
- D** Statement 1: DNA content per cell remains unchanged after meiosis I and is halved after meiosis II.
Statement 2: Ploidy level is halved after both meiosis I and meiosis II.

17. The following diagram shows the development of the blastocyst from a zygote. **P** refers to the cells found within the blastocyst.



Which of the following statements are true?

- I P is the first stem cell produced that will differentiate into all cell types.
- II Both zygote and P are capable of self-renewal.
- III Both zygote and P are genetically identical.
- IV P functions in the growth and development of the organism.

- A I and IV only
- B II and IV only
- C I, II and III only
- D II, III and IV only

- 18.** Embryonic stem cells (ESCs) can be obtained from developing embryos and have the potential to treat diseases.

Both somatic cell nuclear transfer (SCNT) and induced pluripotent stem cells (iPSCs) are techniques to produce ESCs.

What are the advantage(s) of using iPSCs over SCNT?

- I There are no embryos involved.
- II The patient's own somatic cells can be used to produce iPSCs, to overcome immune rejection.
- III There is no danger of iPSCs producing a tumour.

- A** I only
- B** I and II only
- C** II and III only
- D** All of the above

- 19.** The position on a chromosome occupied by a1..... is termed as2.....

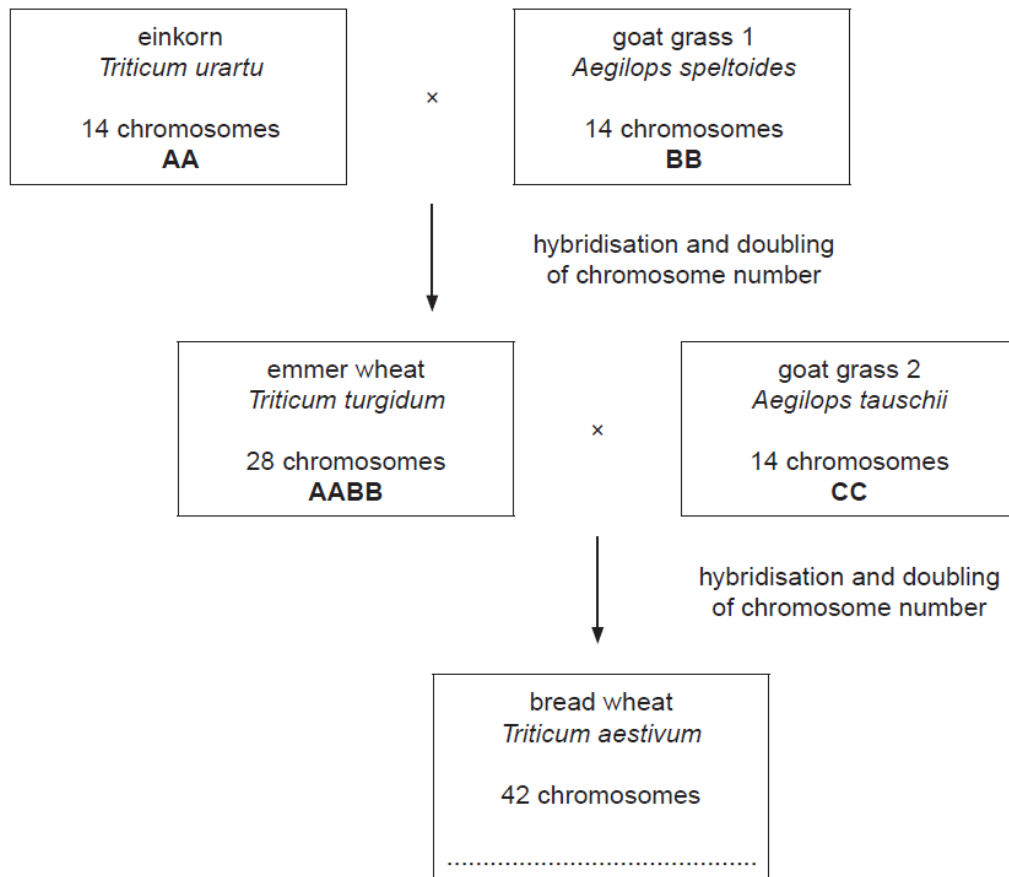
An individual who is considered a3..... contains two different4..... of a5.....

Which of the following correctly completes the sentences?

	1	2	3	4	5
A	DNA	locus	heterozygote	gene	gene
B	gene	centromere	homozygote	gene	trait
C	DNA	centromere	homozygote	allele	trait
D	gene	locus	heterozygote	allele	gene

20. Modern varieties of wheat have developed from numerous hybridisation events between different species of wild grasses. The figure below shows some of the possible steps that are believed to have been involved in the development of bread wheat, *Triticum aestivum*.

The letters, **A**, **B** and **C** represent three different sets of seven chromosomes.



Which of the following letters correctly represent the sets of chromosomes in bread wheat?

- A** ABC
- B** ABCC
- C** AABBCC
- D** AABBCCCC

21. In mice, fur colour is controlled by a gene with multiple alleles. These alleles are listed below in no particular order.

yellow = C^y

agouti = C^a

black = C^b

The following table shows three crosses and the results.

parental phenotype	offspring phenotype
agouti fur crossed with black fur	all agouti fur
yellow fur crossed with black fur	a ratio of one yellow fur to one black fur
yellow fur crossed with yellow fur	a ratio of two yellow fur to one black fur

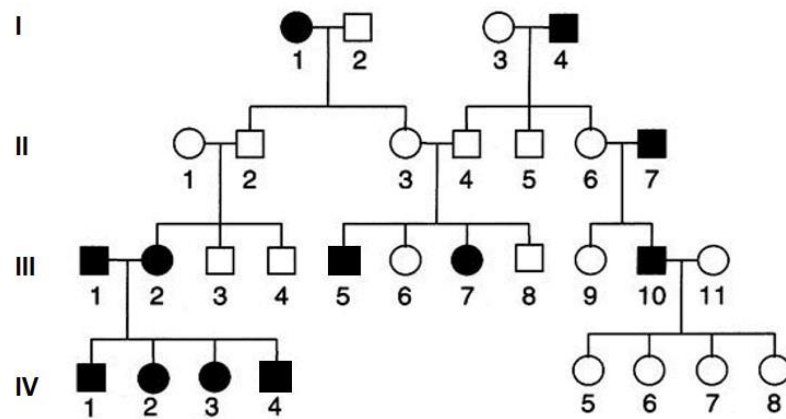
Which of the following can be deduced from the results of the cross?

- I agouti allele, C^a is dominant to black allele, C^b .
- II agouti allele, C^a is dominant to yellow allele, C^y .
- III yellow allele, C^y is dominant to black allele, C^b .
- IV homozygous genotype C^yC^y is lethal and offspring does not survive.

- A** I and II only
- B** I and III only
- C** III and IV only
- D** I, III and IV only

- 22.** Pompe disease is a glycogen storage disease. Individuals with pompe disease shows accumulation of glycogen within the lysosome and subsequent enlargement of the organelle.

The following figure shows the pedigree analysis of two families affected with Pompe disease.



Which of the following conclusion is correct?

- A** Pompe disease is inherited in an autosomal dominant manner as affected individuals III-1 and III-2 have all children who are all affected.
- B** Pompe disease is inherited in an autosomal recessive manner as normal individuals II-1 and II-2 have an affected child, III-2.
- C** Pompe disease is inherited in a X-linked dominant manner as affected individual II-7 have an affected son, III-10.
- D** Pompe disease is inherited in a X-linked recessive manner as normal individual II-3 have both affected son and daughter, III-5 and III-7.

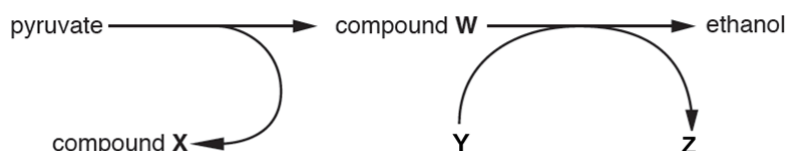
23. 2,4-Dinitrophenol is a banned herbicide. The following statements describes what happens when 2,4-Dinitrophenol is present in cells.

- it is embedded in the inner mitochondrial membrane
- it functions as a protonophore – providing an alternate pathway for H^+ to leak through the inner mitochondrial membrane
- energy from the H^+ gradient is lost as heat

Based on the statements above, which of the following is true about 2,4-Dinitrophenol?

- A** It disrupts the formation of ATP via chemiosmosis, thus no oxygen will be used by the cell.
- B** It disrupts the formation of proton gradient, thus NAD^+ and FAD will not be regenerated.
- C** 2,4-Dinitrophenol is a hydrophilic molecule.
- D** High dose of 2,4-Dinitrophenol will eventually cause the enzymes in the cell to denature, which will then lead to the death of the cell.

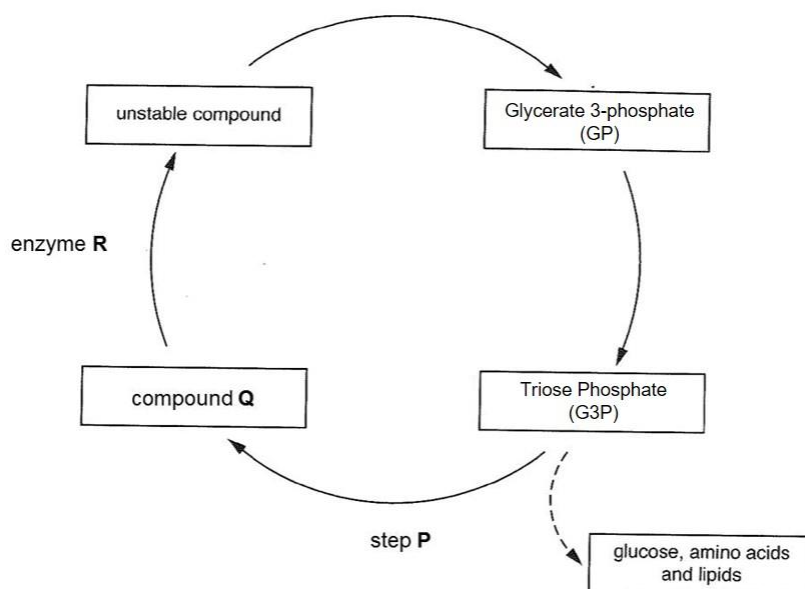
24. The following figure outlines the process of anaerobic respiration in yeast cells.



Which of the following correctly identifies **W**, **X**, **Y** and **Z**?

	W	X	Y	Z
A	acetyl-CoA	carbon dioxide	NADH	NAD^+
B	acetyl-CoA	ATP	NAD^+	NADH
C	ethanal	Carbon dioxide	NADH	NAD^+
D	ethanal	ATP	NAD^+	NADH

25. The following diagram shows the reactions occurring in the Calvin cycle.



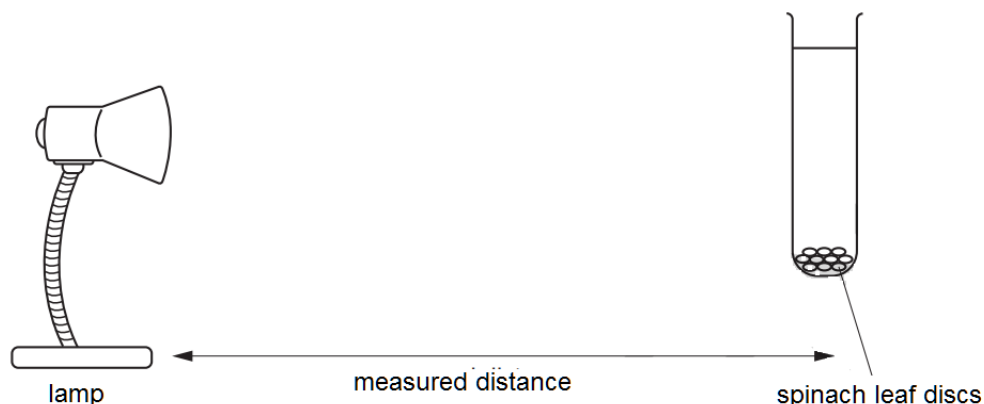
Which of the following correctly identifies **P**, **Q** and **R**?

	P	Q	R
A	regeneration	oxaloacetate	rubisco
B	carbon fixation	RuBP	rubisco
C	regeneration	RuBP	rubisco
D	reduction	rubisco	RuBP

- 26.** A student carried out an investigation on the effect of light intensity on photosynthesis. Several groups of spinach leaf discs were placed in test tubes of water.

The test tubes were then kept in the dark for 30 minutes, the discs all sank to the bottoms of the tubes.

Each tube was placed at a measured distance from a lamp.



As photosynthesis occurs, the build-up of oxygen gas in the leaf discs causes them to rise from the bottom of the tube upwards. The results are shown in the table below,

tube number	distance from lamp (mm)	time taken for five discs to float (s)
1	25	124
2	50	125
3	100	210
4	150	360
5	200	600
6	250	none floated in the time available

Which of the following statements are true?

- I The compensation point occurs after 200 mm.
- II The time taken for discs to rise increases at a constant rate with increasing distance from lamp.
- III At distances between 25 to 50 mm, light intensity is not a limiting factor.
- IV The enzymes involved in photosynthesis are denatured when distance from lamp is equal to or more than 250 mm.

- A** I and III only
- B** I and IV only
- C** I, II, and III only
- D** II, III and IV only

27. Maize, *Zea mays* is a common crop grown in the USA.

The corn borer, *Ostrinia nubilalis*, is an insect pest of maize. The larvae are caterpillars that eat the leaves of the maize plants. The adults can fly. Adult corn borers do not feed on maize plants.

Much of the maize that is grown in the USA has been genetically modified to produce *Bt* toxin, which is lethal to insects that feed on the leaves. However, many populations of the corn borer have now evolved resistance to the *Bt* toxin.

The recessive allele, *r*, of the gene in corn borers confers resistance to *Bt* toxin.

In order to reduce the number of corn borers resistant to *Bt* toxin, farmers are required to grow up to 50% of their maize as non-*Bt* varieties. The non-*Bt* maize is grown in separate areas, called 'refuges', close to the fields of *Bt* maize.

Almost all corn borer larvae feeding on this non-*Bt* maize have the genotypes *RR* or *Rr*. This strategy assumes that, when these become adults, they will interbreed with the adults developing in the *Bt* maize fields.

Which of the following statements are true?

- I Only corn borers with genotype **rr** will turn from larvae into adults in the fields where *Bt* maize is grown.
- II 'Refuges' area ensures that the dominant allele, **R**, stays in the corn borers population.
- III 'Refuges' area drives directional selection for the dominant allele, **R**, in the corn borers population.
- IV The presence of *Bt* resistant corn borer is due to artificial selection.

- A** I and II only
- B** I and III only
- C** II and III only
- D** I, II and IV only

28. Cystic fibrosis is caused by a mutation to the CFTR gene. The CFTR gene codes for a transmembrane channel involved in the transport of chloride ions. Mutant alleles leads to production of either non-functioning channels or no protein product at all. The lack of functioning CFTR transmembrane channel can lead to production of thick and sticky mucus that can affect the function of the lungs and digestive system. Individuals with cystic fibrosis tend to have a shorter-than-normal life span.

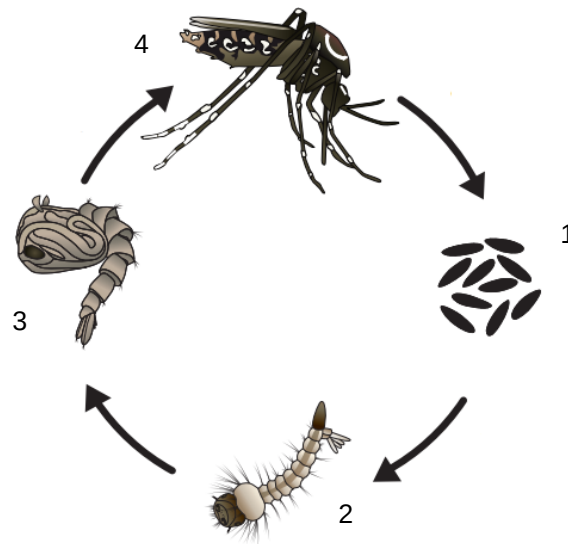
Cholera is a disease caused by the infection of the bacterium *V. cholerae*. Upon infection, *V. cholerae* secretes a toxin that dysregulates the CFTR transmembrane channel leading to imbalance of ion and water in the gastrointestinal tract. This causes symptoms like diarrhoea, severe conditions can lead to dehydration and death.

Individuals with cystic fibrosis appears to be immune to the symptoms of Cholera.

Based on the information provided, which of the following statement is true?

- A** Overtime, in a population exposed to *V. cholera*, the allele frequency for the mutant CFTR allele will drop to 0% as individuals with cystic fibrosis have a shorter-than-normal life span.
- B** Individuals with cystic fibrosis will be at selective advantage in populations exposed to *V. cholerae* due to the lack of functioning CFTR transmembrane channel.
- C** The CFTR mutant allele remains in population exposed to *V. cholerae* due to heterozygote protection.
- D** The CFTR mutant allele remains in population exposed to *V. cholerae* due to heterozygote advantage.

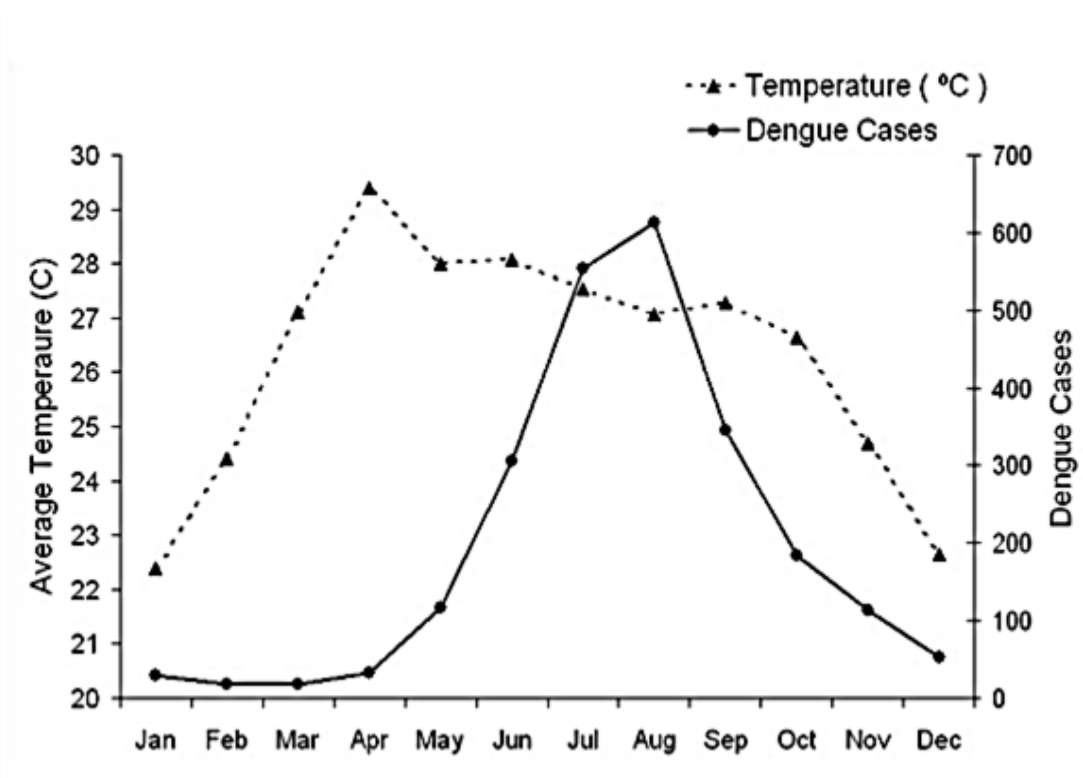
29. The life-cycle of *Aedes aegypti* is shown below.



Which of the following statement is true?

- A Stage 1 can withstand prolonged periods of dry conditions.
- B Only two out of the four stages occur in water.
- C Stage 3 is infected by the dengue virus before stage 4 infects human hosts.
- D Both males and females of stage 4 can act as vectors for the dengue virus.

30. The following graph shows how changes in temperature affects the average month incidence of dengue cases in Chiang Rai, Thailand from January to December, 2004.



Which of the following statement(s) accounts for the trend observed?

- I Increase in temperature causes increase in metabolic activity mosquitoes, leading to faster growth and maturation rate.
- II Mosquitoes are not able to tolerate average temperature of beyond 27°C.
- III The delay between rise in dengue cases and temperature increase is because time is needed for mosquitoes to grow, mature and act as vector for dengue.
- IV Dengue cases are not affected by temperature as the two graphs does not show a correlation.

- A IV only
- B I and II only
- C I and III only
- D I, II and III only

End of Paper